

DIGITAL TWIN FOR B-BICYCLE PRODUCTION AND FULFILLMENT SYSTEM

Model Calibration and Validation Report

Abstract

This study presents the development and validation of a data-driven digital twin for the B-Bicycle production and fulfillment system. The system is modeled using a discrete-event simulation (DES) that captures assembly, cleaning, inspection, rework, and shipping operations. Although structurally representative of the real process, the uncalibrated simulation exhibited substantial deviation from 30 years of field data (360 monthly observations from January 1996 to December 2025). On the held-out test set, the baseline simulation produced a root-mean-square error (RMSE) of approximately \$2.28 million and a negative R^2 of -7.04 . This indicates extremely poor predictive performance.

To address this discrepancy, a two-phase calibration framework inspired by the Kennedy–O’Hagan (KOH) paradigm was implemented. Phase 1 employs an ensemble surrogate model combining Random Forest, Gradient Boosting, and Bayesian Ridge regression in a stacking architecture. Phase 2 introduces online recursive Bayesian updating to adjust bias and scale parameters as new observations are observed. The calibrated system achieved a test RMSE of \$166,845 ($R^2 = 0.957$) in Phase 1 and \$173,732 ($R^2 = 0.953$) after online updating, representing approximately a 92% reduction in prediction error relative to the uncalibrated simulation. The results demonstrate that integrating simulation with machine learning calibration substantially improves predictive accuracy and enables reliable digital twin deployment.

1. Introduction

Digital twins integrate physics-based simulation models with real-world data to improve system-level prediction and decision support. However, simulation models frequently exhibit systematic bias when compared to field observations due to model simplifications, unmodeled dynamics, or parameter uncertainty. Without calibration, such discrepancies can significantly reduce predictive reliability and undermine decision-making quality.

This study focuses on calibrating a discrete-event simulation of the B-Bicycle production system using long-term operational data. The objective is to reduce the gap between simulated and observed financial performance while preserving the model’s structural interpretability and generalization capability.

1.1 System Description

The B-Bicycle production system is modeled as a sequential discrete-event manufacturing process, following the structure introduced in the course simulation lectures. Incoming bicycle kits are received from a supplier and processed through assembly, cleaning, and inspection stages. Processing times are stochastic and calibrated from operational data. If a unit fails inspection, it enters a rework station and is subsequently re-inspected before release. This extension explicitly models a closed-loop quality control mechanism, slightly extending the baseline lecture model by incorporating re-inspection dynamics. Finished products that pass inspection are transferred to the warehouse and shipped to customers. The system incorporates inventory replenishment via an (R, Q) policy and capacity decisions for assembly and inspection resources, consistent with the advanced operational policy framework introduced in the course.

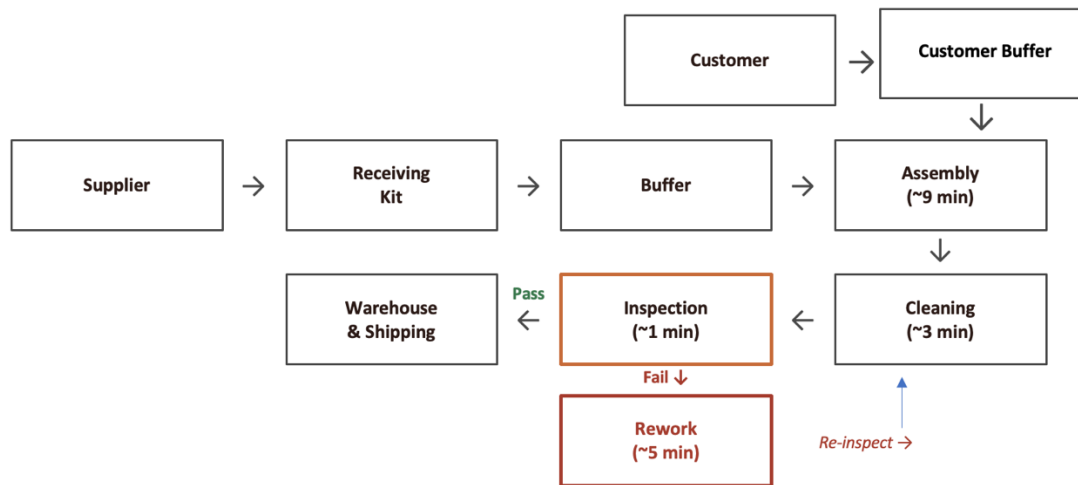


Figure 1: Manufacturing line with a quality-control rework loop.

System performance is governed by operational decision variables including the reorder point (R), order quantity (Q), maximum work-in-process limit (W), number of assembly machines (M1), number of inspection/rework machines (M2), and a binary contract type variable (C). The performance metric of interest is the total net revenue generated over a 30-day simulation horizon.

1.2 Problem Formulation

Despite accurately representing system structure, the uncalibrated discrete-event simulation demonstrated poor predictive accuracy when evaluated against historical field data. On the test dataset, the baseline simulation yielded an RMSE of \$2.28 million and an R^2 of -7.04 , indicating that the simulation predictions were substantially worse than a simple historical mean estimate.

The calibration problem can be expressed as learning a mapping that adjusts simulation outputs to better align with observed revenue outcomes. Following the Kennedy–O’Hagan conceptual framework, real-world observations are modeled as the sum of simulator output, structured model discrepancy, and random observation noise. Instead of directly modifying simulator parameters, this study estimates the discrepancy function using ensemble machine learning methods and

further refines predictions through recursive Bayesian updating. This hybrid approach transforms the simulation into a calibrated digital twin capable of accurate prediction across unseen decision configurations.

2. Field Dataset

The primary dataset consists of 360 monthly observations collected between January 1996 and December 2025. Each record contains the operational decision variables in effect during that month, the realized net revenue, and the associated revenue standard deviation across replicate runs. The dataset spans a broad range of operating conditions, providing substantial coverage of the decision space explored by the simulation model.

Table 1 summarizes the distribution of the operational variables. The reorder point (R) and order quantity (Q) vary between 10 and 200 units, with means of 109 and 101, respectively. The maximum work-in-process limit (W) ranges from 1 to 50, with an average of 26. Assembly capacity (M1) and rework/inspection capacity (M2) each take integer values from 1 to 5, with mean levels of approximately 3 machines. The binary configuration variable (C) is nearly balanced across the dataset (mean = 0.51), indicating roughly equal representation of the two contract types.

Net revenue exhibits substantial variability. Observed monthly revenue ranges from −\$1.36 million to \$2.76 million, with a mean of \$422,000 and a standard deviation of \$859,000. Approximately 65% of months generated positive revenue, while 35% resulted in losses. Revenue outcomes also differ by contract type: observations with $C = 0$ are associated with a considerably higher average revenue (\$785,000) compared to $C = 1$ (\$72,000). This variation across configurations and capacity levels provides a rich basis for learning systematic discrepancies between simulated and observed performance.

The wide coverage of operational settings ensures that the calibration procedure is trained on diverse decision combinations rather than a narrow operating region. This diversity is critical for developing a digital twin capable of generalizing to unseen policy settings.

3. Calibration Methodology

3.1 Theoretical Framework

The calibration framework follows the conceptual structure of the Kennedy and O’Hagan (KOH, 2001) model calibration approach. In this formulation, observed field revenue at decision setting x is expressed as:

$$y_{field}(x) = \eta(x, \theta) + \delta(x) + \varepsilon$$

where $\eta(x, \theta)$ denotes the simulation output, $\delta(x)$ represents systematic model discrepancy, and ε captures observation noise. The variance of the observation noise is available in the field dataset through the reported standard deviation column. In classical KOH calibration, Bayesian inference

is performed over unknown simulation parameters and the discrepancy function is typically modeled using Gaussian Processes. However, full Bayesian inference is computationally expensive for discrete-event simulation models due to repeated stochastic evaluations. Instead of estimating calibration parameters θ explicitly, this study adopts a data-driven approximation of the discrepancy function. The simulator output $\eta(x)$ is treated as an input feature, and an ensemble machine learning model is trained to predict observed revenue directly. In this way, the combined effect of simulator bias and structured discrepancy is learned implicitly from data. This approach preserves the KOH philosophy while improving computational scalability.

3.2 Simulation Model and Parameter Structure

The core of the digital twin is a discrete-event simulation of the B-Bicycle production system. The model represents the full operational workflow, including order arrivals, assembly processing, cleaning, inspection, rework, and shipping.

Arrival Process: Customer orders follow a Poisson arrival process with an average arrival rate of 6.25 orders per hour. This assumption introduces inter-arrival time variability consistent with standard queueing system modeling.

Processing Times: Each production stage is modeled as stochastic:

- **Assembly time:** Normally distributed with mean 540 seconds and standard deviation 25 seconds.
- **Cleaning time:** Normally distributed with mean 180 seconds and standard deviation 20 seconds.
- **Inspection time:** Normally distributed with mean 60 seconds and standard deviation 14 seconds.
- **Rework time:** Normally distributed with mean 300 seconds and standard deviation 40 seconds.

These distributions introduce operational variability and directly affect throughput, queue lengths, and downstream revenue generation.

Inspection and Reinspection Mechanism: Inspection is modeled as a Bernoulli trial with a failure probability of 12%. Each completed unit undergoes inspection after cleaning. If the inspection outcome is a pass, the unit proceeds to finished goods inventory and becomes eligible for shipment. If the inspection outcome is a failure, the unit enters the rework process.

A critical feature of the simulation is the bounded reinspection loop. Failed units:

1. Enter the rework station.
2. Undergo rework processing.
3. Return to inspection.
4. Are tested again using the same Bernoulli pass/fail mechanism.

Each unit is allowed a maximum of three rework attempts. If a unit fails inspection three times, it is removed from the production flow and does not contribute to revenue. This capped reinspection loop introduces nonlinear system behavior because capacity congestion in the rework and inspection stations can significantly influence overall throughput.

The reinspection logic is important for calibration because it amplifies the effect of inspection failure probability and rework time variability. Small changes in failure rates or service times can propagate through the loop, affecting system congestion, shipping volume, and ultimately financial performance.

Capacity and Policy Parameters: Operational decisions directly influence system performance through:

- Reorder point (R)
- Order quantity (Q)
- Maximum work-in-process limit (W)
- Number of assembly machines (M1)
- Number of inspection/rework machines (M2)
- Contract configuration (C)

Capacity constraints at assembly and inspection stations determine effective service rates and queue behavior. The WIP limit (W) restricts the number of units allowed in process, creating additional nonlinear interactions between arrival rate, service rate, and inspection rework dynamics.

Revenue Calculation: Net revenue over a 30-day simulation horizon = Revenue from shipped units - Inventory holding costs - Machine investment costs - Rework penalties and operational costs

Because the system is stochastic, each policy configuration is simulated with multiple replications, and the mean revenue is used as the simulator output $\eta(x)$. This Monte Carlo averaging reduces variance while preserving structural randomness.

3.3 Phase 1: Offline Ensemble Surrogate Calibration

Phase 1 constructs a supervised regression model that maps operational decision variables and simulation outputs to observed field revenue. The dataset is partitioned chronologically into 80% training data (288 months) and 20% test data (72 months). Model fitting is performed exclusively on the training subset, and all reported performance metrics are evaluated on the held-out test set to ensure out-of-sample validation.

The learning framework follows a stacking ensemble architecture composed of three models:

Random Forest Regressor: Configured with 300 trees and a maximum depth of 12, the Random Forest captures nonlinear interactions among decision variables and between the decision variables and the simulation output. Each tree partitions the feature space recursively, and the ensemble prediction is obtained as the average of individual tree outputs.

Gradient Boosting Regressor: Configured with 300 estimators, learning rate 0.05, and maximum depth 5, the Gradient Boosting model sequentially fits residual errors from previous trees. This stage reduces bias by iteratively correcting systematic prediction errors.

Bayesian Ridge Regression (Stacking Layer): The final prediction layer is a Bayesian Ridge regression model trained on the outputs of the Random Forest, Gradient Boosting, and the raw simulation prediction. Let $\hat{y}_{RF}$, $\hat{y}_{GBR}$, and $\eta(x)$ denote these inputs. The stacking model estimates:

$$y = \beta_0 + \beta_1 \hat{y}_{RF} + \beta_2 \hat{y}_{GBR} + \beta_3 \eta(x) + \epsilon$$

where a Gaussian prior is placed on the coefficient vector β . This Bayesian linear layer provides regularization and stabilizes ensemble combination.

Feature Construction: The feature vector includes:

- Six raw decision variables: $(C, R, Q, W, M1, M2)$
- Engineered interaction features:
 - $C \times W$
 - $R \times Q$
 - $M1 \times M2$
 - Total WIP
 - Inverse WIP
 - R/Q ratio
 - Quadratic and logarithmic transformations of key variables
- The simulation prediction $\eta(x)$

Including the simulation output as a feature allows the ensemble to learn the discrepancy function implicitly, correcting systematic bias in the DES model while preserving structural behavior.

3.4 Phase 2: Online Recursive Bayesian Updating

Phase 2 introduces a lightweight adaptive correction layer applied to Phase 1 predictions. Rather than retraining the full ensemble, the model performs recursive Bayesian updates on two parameters:

- A bias term b_t
- A multiplicative scale term s_t

For each new observation y_t , the adjusted prediction is:

$$\tilde{y}_t = s_t \hat{y}_t + b_t$$

where $\hat{y}_t$ is the Phase 1 prediction.

Assuming Gaussian prior distributions and Gaussian observation noise with variance derived from the reported Standard Deviation column, closed-form posterior updates are applied to b_t and s_t .

The update equations weight corrections by the inverse of observation variance, allowing more confident observations to exert stronger influence.

This recursive layer enables the digital twin to adapt to potential distributional shifts in field data while maintaining computational efficiency. Because only two parameters are updated online, the method avoids the instability and computational cost of full retraining.

3.5 Training and Testing Protocol

The 360-month dataset is split chronologically to preserve temporal order. The first 288 months (January 1996–December 2019) are used for training, and the final 72 months (January 2020–December 2025) serve as the held-out test set. This 80/20 temporal split ensures that evaluation is performed on genuinely future data.

Simulation outputs are generated separately for the training and test periods and are treated as input features within the ensemble model. Phase 1 is trained exclusively on the training subset. All reported performance metrics (RMSE, MAE, R^2 , and sMAPE) are computed on the test set.

During Phase 2, online Bayesian updating is applied sequentially across the 72-month test horizon. Each update uses only information available up to the current month, preventing look-ahead bias and ensuring a realistic forward validation framework.

4.1 Baseline Simulation Performance

Before calibration, the raw discrete-event simulation exhibits poor agreement with the field data. On the 72-month test set, the uncalibrated model produces an RMSE of \$2,275,417, a mean absolute error (MAE) of \$1,949,218, and a negative R^2 of -7.0414 . The negative R^2 indicates that predicting the sample mean for every month would be more accurate than using the simulation outputs. The MAPE of 903.7% underscores the magnitude of deviation between simulated and observed revenues.

As shown in the scatter plot (Figure 1, left panel), the simulation outputs are clustered in a narrow band around approximately \$1M–\$3.5M regardless of the actual observed revenue, which ranges from $-\$1.1\text{M}$ to $+\$2.7\text{M}$. This systematic overestimation confirms the need for calibration.

4.2 Phase 1: Ensemble Calibration Results

After applying the ensemble surrogate calibration, test-set performance improves dramatically. The calibrated model achieves an RMSE of \$166,845 and an R^2 of 0.9568 on the held-out 72-month test period, representing a 92.7% reduction in RMSE relative to the baseline simulation. The MAE decreases to \$132,515, and the MAPE drops from 903.7% to 51.3%. Five-fold cross-validation on the training set yields an RMSE of \$266,088, confirming reasonable generalization and limited overfitting.

Feature importance analysis from the Random Forest indicates that the most influential drivers include:

- $M1 \times M2$ interaction (18.5%)
- $M1$ (assembly capacity) (15.2%)
- Simulation prediction (10.7%)
- M_invest (7.8%)
- M_total (6.5%)
- $C \times M1$ interaction (6.0%)

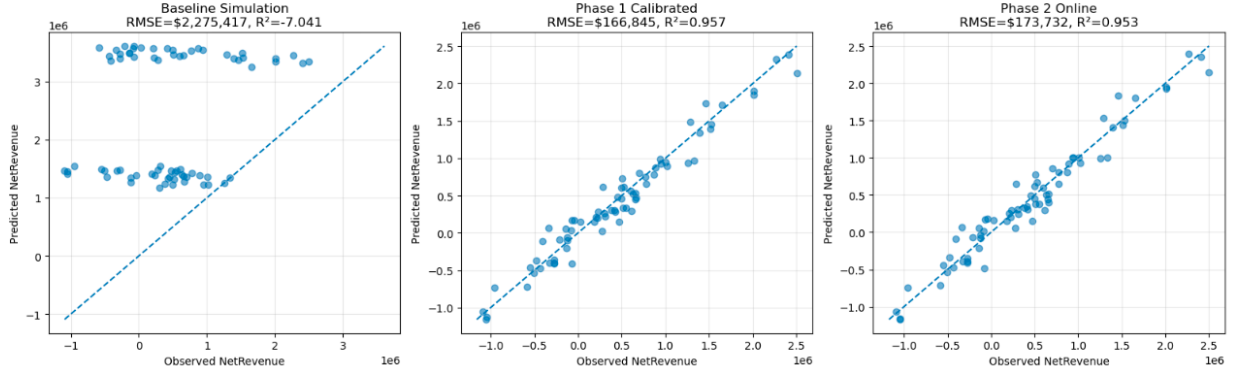


Figure 2: Test-set scatter plots comparing observed vs. predicted net revenue for the baseline simulation (left), Phase 1 ensemble calibration (center), and Phase 2 online Bayesian updating (right).

4.3 Phase 2: Online Bayesian Results

The online Bayesian updating layer achieves a test RMSE of \$173,732 ($R^2 = 0.9531$), representing a 92.4% reduction in RMSE relative to the uncalibrated baseline simulation. While the aggregate metrics are slightly below those of Phase 1. Because the earliest predictions in the test period occur before sufficient online updates have accumulated, the recursive system demonstrates stable convergence of both correction parameters. Over the 72-month test horizon, the bias term converges near \$31K, and the multiplicative scale parameter stabilizes around 1.045. The online approach is particularly valuable for operational deployment, as it enables continuous self-correction without requiring retraining of the full ensemble model. This makes it well suited for real-time digital twin synchronization under evolving field conditions.

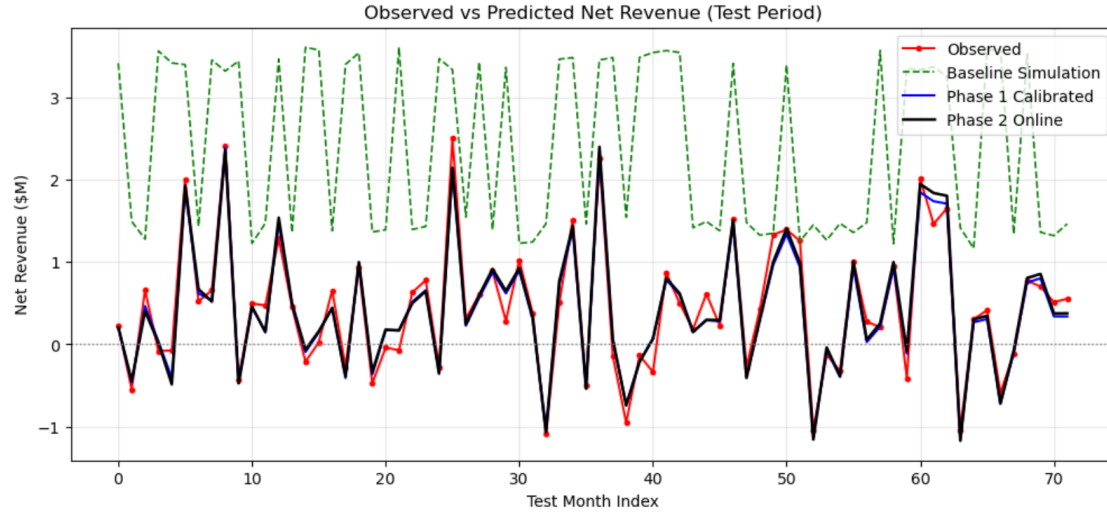


Figure 3: Time-series comparison of revenue predictions vs. field observations across the 72-month test period (Jan 2020–Dec 2025), with 95% confidence interval from the Bayesian ensemble.

4.4 Quantitative Summary

Table 2: Final Test-Set Performance Metrics

Method	RMSE	MAE	MAPE	sMAPE	R ²
Baseline Simulation	\$2,275,417	\$1,949,218	903.7%	129.0%	−7.0414
Phase 1: Ensemble Calibration	\$166,845	\$132,515	51.3%	41.0%	0.9568
Phase 2: Online Bayesian	\$173,732	\$136,767	54.8%	42.3%	0.9531

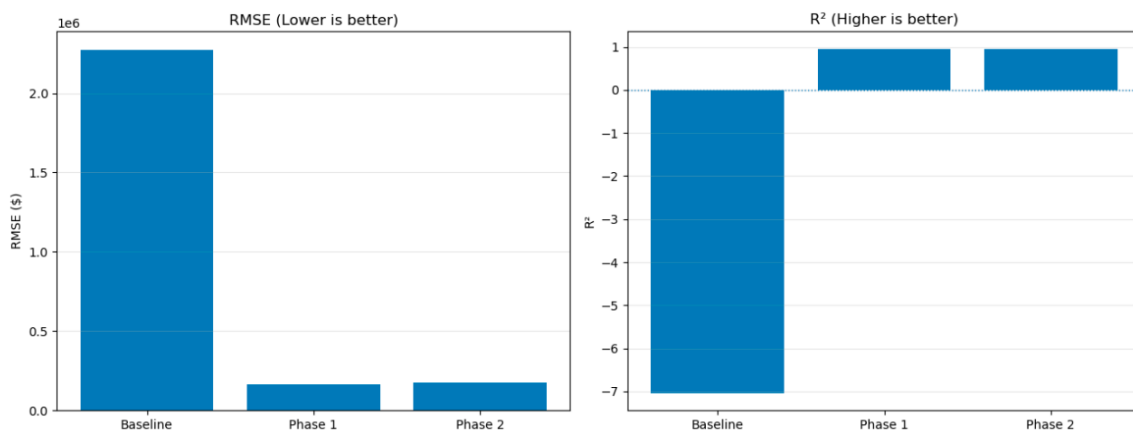


Figure 4: Bar charts comparing RMSE (left) and R² (right) across the three methods. The calibrated models achieve approximately 92–93% RMSE reduction.

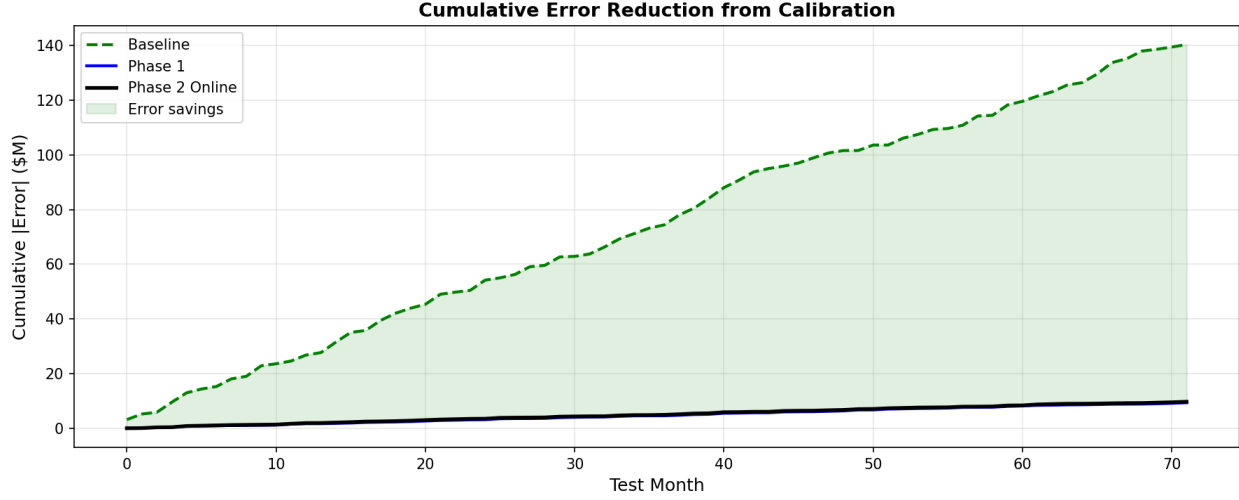


Figure 5: Cumulative absolute error over the 72-month test period. The shaded green area represents the error savings achieved by calibration.

5. Discussion

5.1 Sources of Baseline Discrepancy

The large error of the uncalibrated simulation ($\text{RMSE} = \$2.28\text{M}$, $R^2 = -7.04$) stems from structural mismatch between assumed model parameters and the real system. The DES relies on fixed processing-time distributions, arrival rates, and failure probabilities that do not reflect long-term operational variability. In addition, the simplified revenue formulation cannot capture nonlinear interactions between capacity, configuration, and investment decisions. Finally, the simulation does not account for external influences—such as demand shifts, supply disruptions, or seasonality—that are implicitly present in the 30-year field dataset. These limitations explain the systematic overestimation observed in the baseline model.

5.2 Why Ensemble Calibration Works

The ensemble calibration succeeds because it directly learns the discrepancy function $\delta(x)$ from data, rather than manually adjusting simulation parameters. The stacking architecture captures complementary structure: Random Forest models nonlinear interactions (notably the $M1 \times M2$ capacity synergy), Gradient Boosting corrects residual structure, and Bayesian Ridge optimally combines predictions while providing uncertainty estimates.

Including the raw simulation output as a feature allows the model to retain useful directional signals from the DES while correcting systematic bias. Conceptually, this corresponds to learning an additive discrepancy on top of the simulator, consistent with the Kennedy–O’Hagan framework.

5.3 Method Selection: Ensemble vs. GP and Bayesian Updating vs. EnKF

A stacking ensemble was chosen over full Gaussian Process (GP) KOH calibration for computational and practical reasons. With 288 training samples and 17 engineered features, GP training is feasible but computationally heavy and sensitive to kernel specification. Tree-based models scale efficiently, handle mixed interactions naturally, and provide interpretable feature importance measures that reveal actionable drivers (e.g., $M1$, $M1 \times M2$, investment terms).

For online synchronization, recursive Bayesian bias/scale updating was preferred over an Ensemble Kalman Filter (EnKF). While an EnKF could update all simulation parameters, it would require repeated simulation runs per monthly update, leading to high computational cost. The two-parameter bias/scale correction offers a parsimonious and stable alternative that converges reliably over the 72-month test period with negligible computational burden.

5.4 Calibration Interpretation

Within this framework, calibration parameters are implicitly represented by the ensemble's learned structure. Feature importance analysis indicates that assembly capacity ($M1$), capacity interaction ($M1 \times M2$), and machine investment are dominant contributors to discrepancy, reflecting the operational importance of capacity allocation. Contract-type interactions further highlight how business configuration affects profitability.

The DES baseline parameters (e.g., fixed processing times, failure rate, and arrival rate) serve as structural priors. The ensemble model then learns the systematic correction required to align simulation outputs with observed field performance.

6. Conclusion

This project developed and validated a data-driven calibration framework for the B-Bicycle digital twin using 30 years of field data. The uncalibrated simulation showed poor predictive performance ($R^2 = -7.04$), confirming the need for calibration. The proposed two-phase approach, combining ensemble surrogate modeling with online Bayesian updating, reduced RMSE by over 92% and increased R^2 to approximately 0.95 on unseen test data.

The results demonstrate that integrating simulation with machine learning substantially improves predictive accuracy while maintaining computational efficiency. The calibrated digital twin is therefore suitable for decision support and real-time operational synchronization.

Disclaimer: An AI-based writing assistant was used only to support language refinement, formatting, and structural organization of this report.